

AG35-CET&AG35-EUT

QuecOpen Reference Design

LTE Standard Module Series

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About the Document

Revision History

Version	Date	Author	Description
-	2023-09-12	Bob WANG	Creation of the document
1.0.0	2023-09-12	Bob WANG	Preliminary
			First official release
			1. Added the applicable module AG35-EUT.
			2. Updated pin 127 from ANT_DRX to ANT_DIV.
			3. Updated the VBAT current requirement (Sheets 1, 2 and 5).
			4. Deleted test points reserved for PWRKEY and VBAT pins (Sheets 1 and 14).
			5. Updated notes in module interface design (Sheet 3):
			● Updated the note about using GNSS_TXD/RXD;
			● Added a note about adding 33 pF capacitors to prevent RF interference;
			● Added a note about no current sink on the module's pins before the module turns on.
1.0	2025-06-12	Hale LI/ Vigoss YU	6. Updated the recommended VBAT operating voltage (Sheet 5).
			7. Updated the parasitic capacitance requirement of the TVS array (Sheet 6).
			8. Updated the pull-up resistor requirement of I2C interface of the codec (Sheet 7).
			9. Updated the antenna interface design (Sheet 10):
			● Updated the capacitance value of DC-blocking capacitors and added the related note;
			● Updated the reference circuit of GNSS antenna.
			10. Updated the reference design of level-shifting circuit–transistor solution (Sheet 11).
			11. Updated the note about power supply of USB_VBUS

and added a note about its control circuit (Sheet 11).

12. Updated the eMMC application design and deleted supported eMMC models (Sheet 12).
 13. Updated the reference design of forced download mode (Sheet 14).
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1 Reference Design

1.1. Introduction

This document provides the reference design for Quectel AG35-CET and AG35-EUT modules in QuecOpen® solution, including the designs of block diagram, power supply, module interfaces, (U)SIM interface, audio codec, analog audio interface, antenna interfaces, UART interface, USB interface, eMMC application, as well as the design between AG35-CET/AG35-EUT and FC30R (Wi-Fi module).

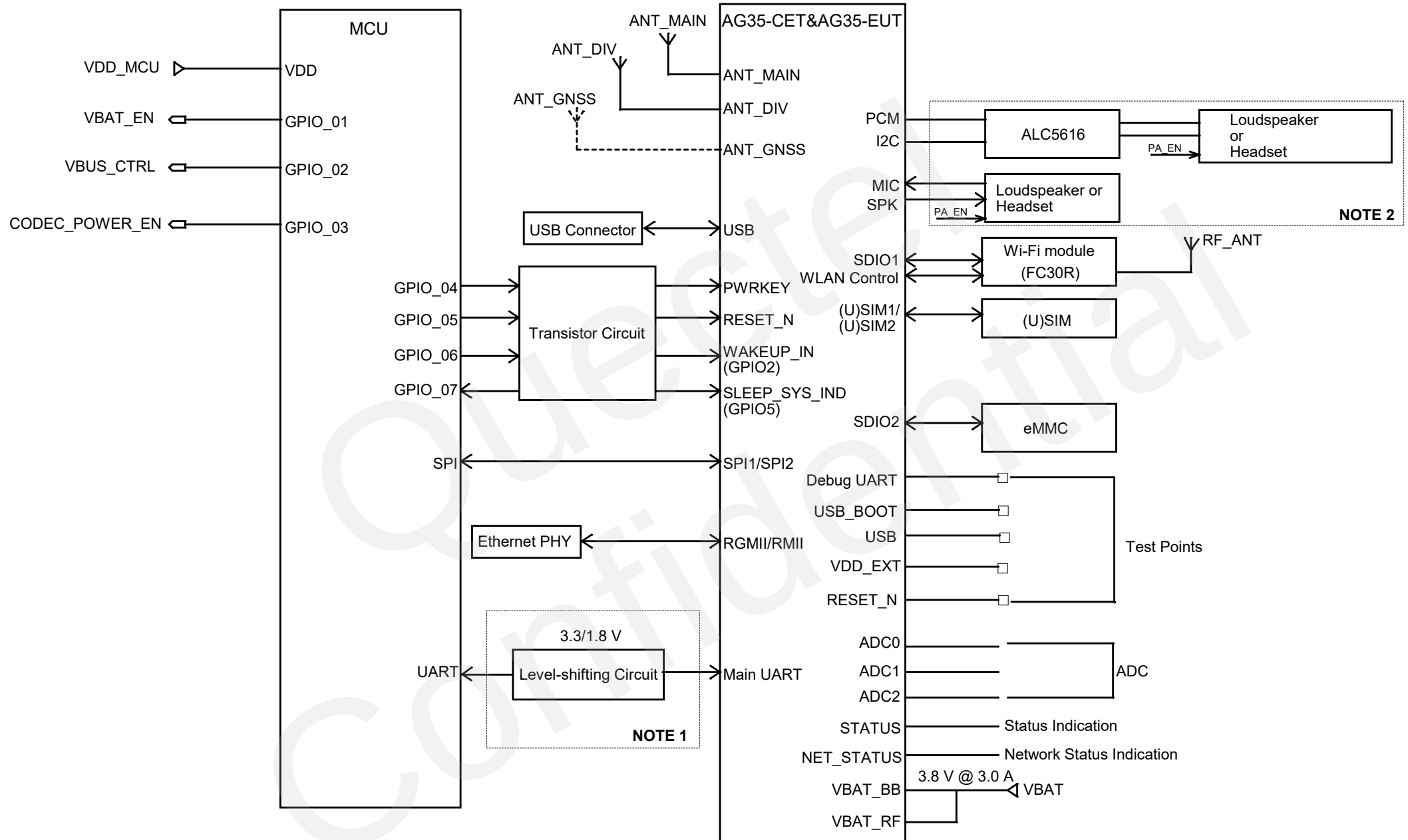
1.2. Schematics

The schematics illustrated in the following pages are provided for your reference only.

NOTE

It is required to confirm the applicability and price from the supplier about the IC involved in the reference design.

Block Diagram

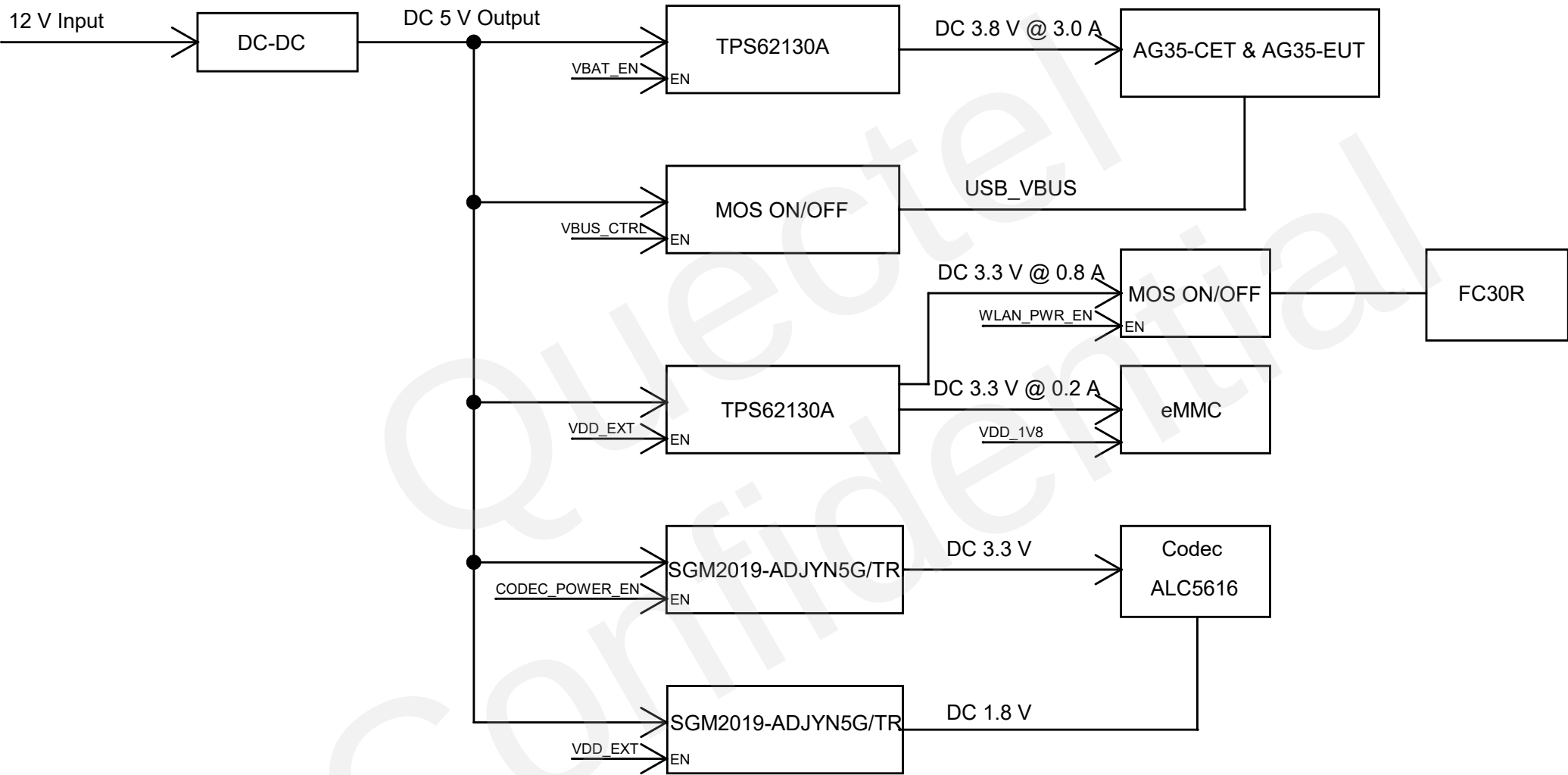


NOTE:

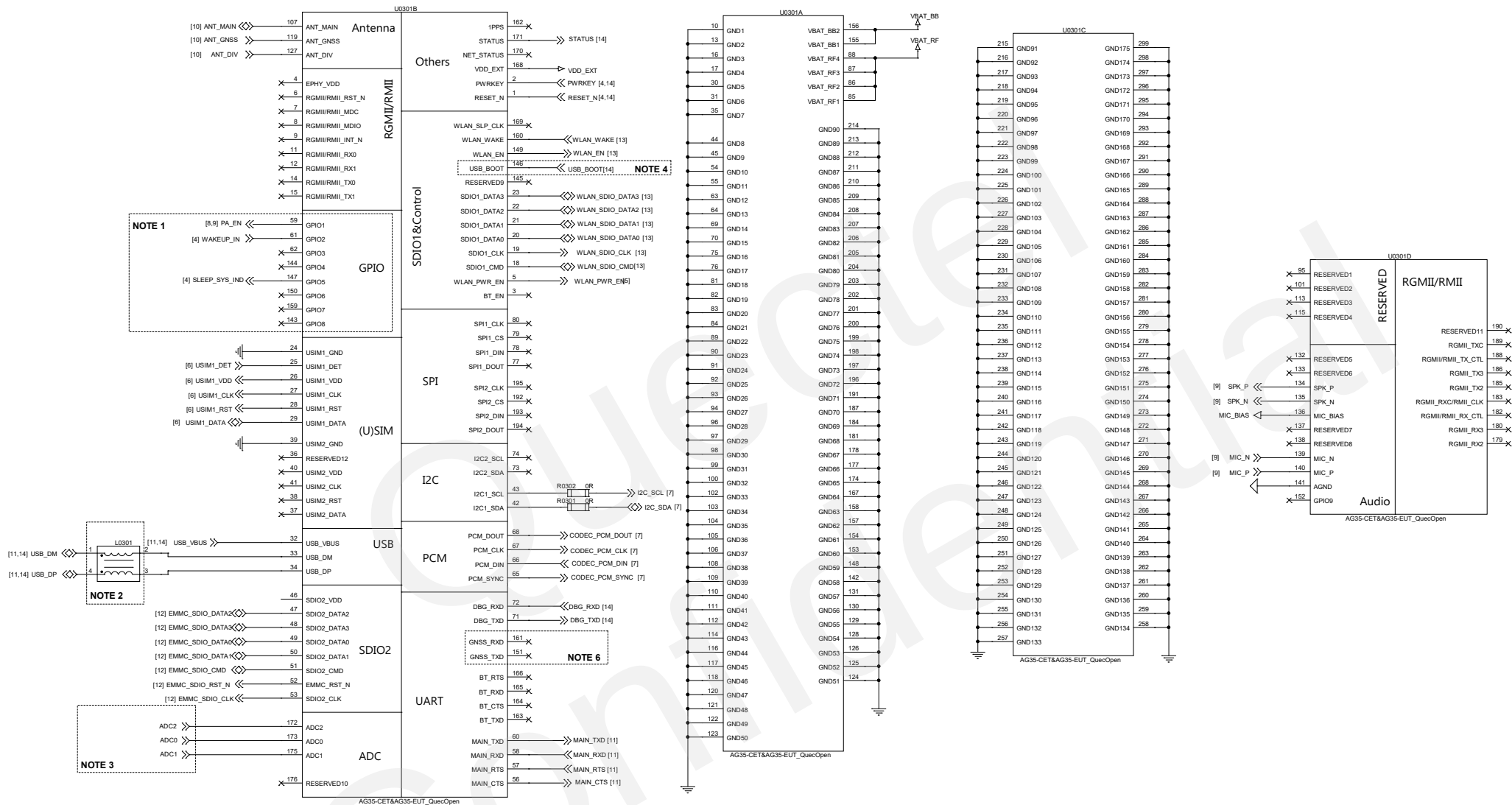
1. A transistor circuit or a voltage-level translator is recommended if the voltage level of MCU is not 1.8 V.
2. You can only select either the analog audio interface or the digital audio interface. They cannot be used simultaneously. The digital audio interface is recommended.

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Power System Block Diagram



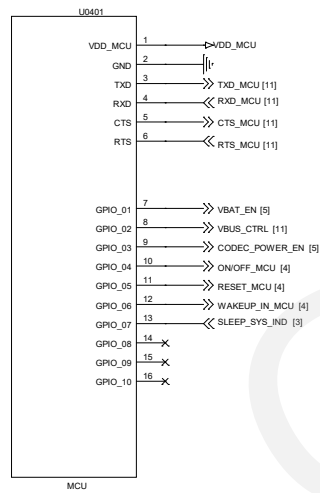
Module Interface Design



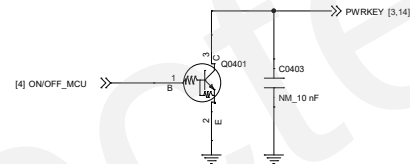
NOTE:

1. It is recommended to reserve a 10 nF capacitor near the module for each GPIO.
2. It is recommended to add a common-mode choke in series between USB connector and the module to filter out EMI. Place the common-mode choke as close to the module as possible.
3. The input voltage range of the ADC pins is 0–VBAT_BB, and the ADC0/ADC1 interface is preferred.
4. Do not pull up USB_BOOT (pin 146) to a high level before the module is turned on successfully, otherwise the module cannot be turned on properly.
5. All GND pins should be connected to the ground, and keep unused and RESERVED pins unconnected.
6. If the module has a built-in GNSS, keep the GNSS_TXD/RXD pin open; if the module does not have a built-in GNSS, the GNSS_TXD/RXD pin can be used for GNSS data communication when an external GNSS is connected.
7. If PCM_CLK (pin 67), SDIO2_CLK (pin 53), I2C_SCL (pins 43 and 74), SPI_CLK (pins 80 and 195) and SDIO1_CLK (pin 19) pins are unused, it is recommended to reserve a 33 pF capacitor near each of these pins to prevent RF interference.
8. Ensure that there is no current sink on the module's pins before the module turns on.

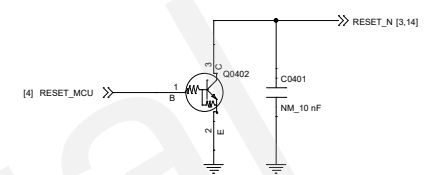
MCU Interface Design



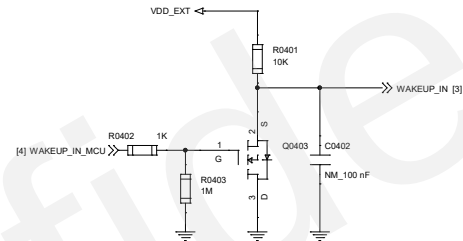
Module On/Off Control



Module Reset Control



Module Wake-up Control



NOTE:

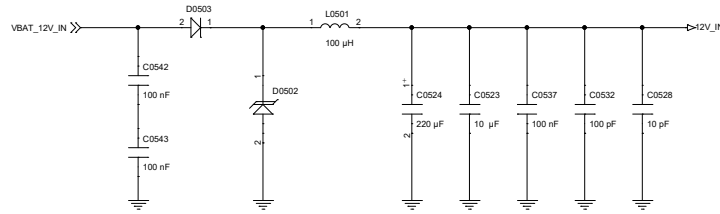
1. U0401 represents your MCU. The power domain of GPIOs of the module is 1.8 V. If the power domain of U0401 is also 1.8 V, then the related level-shifting circuit is not needed.
2. It is recommended to select GPIO pins that are at a low level by default as the control pins for PWRKEY and RESET_N of the module.

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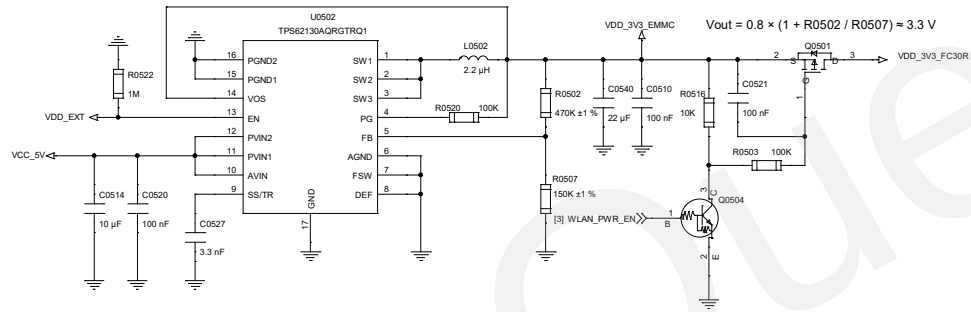
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Power Supply Design

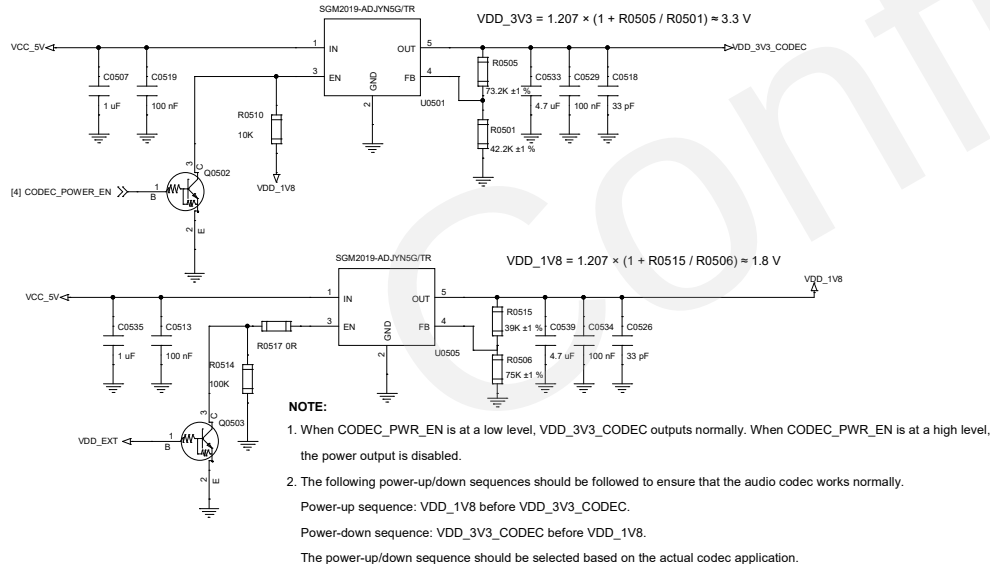
Car Battery 12 V Input



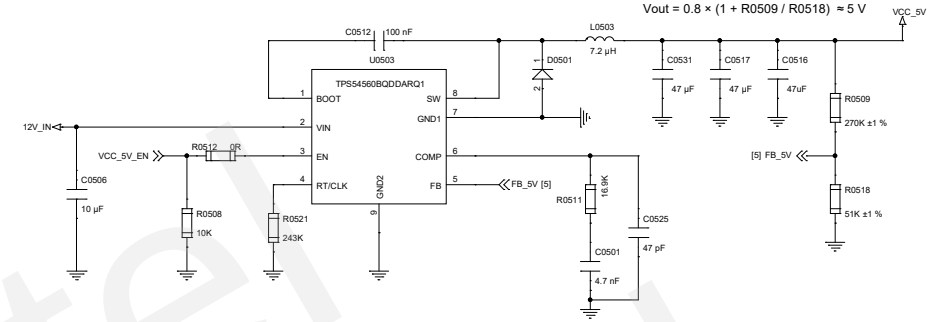
3.3 V Power Supply for FC30R and eMMC



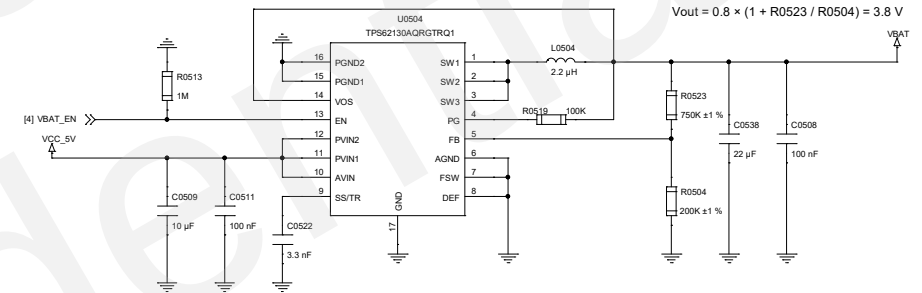
Power Supply for Codec



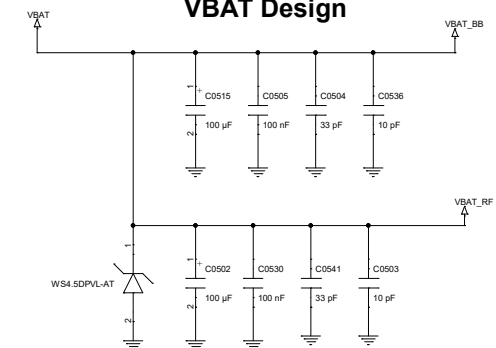
12 V to 5.0 V DC-DC Design



5.0 V to 3.8 V DC-DC Design



VBAT Design



NOTE:

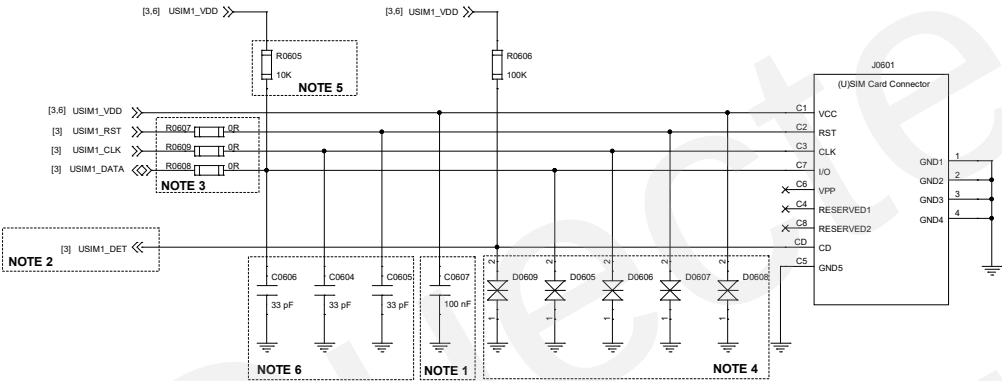
- The VBAT power supply should be able to provide sufficient current of at least 3 A for the module.
- VBAT should be routed in star configuration to VBAT_BB and VBAT_RF pins.
- The recommended operating voltage of VBAT is 3.5-4.3 V.

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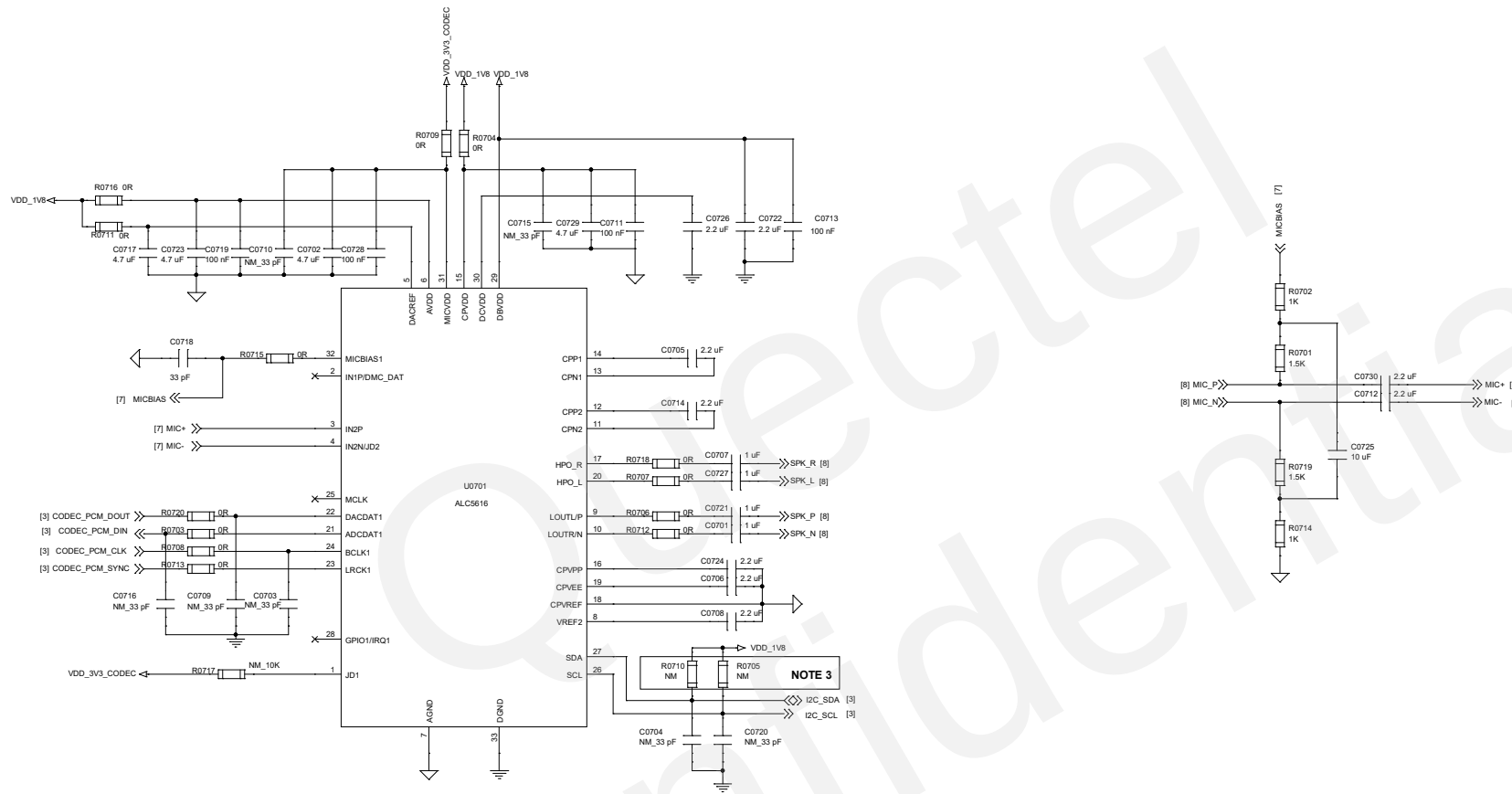
(U)SIM Interface Design

(U)SIM1 Interface



- NOTE:**
1. The decoupling capacitor of USIM1_VDD should be not more than 1 μ F and must be placed to (U)SIM card connector.
 2. The module provides an input pin (USIM1_DET) to detect the presence or absence of a (U)SIM card.
 3. R0607–R0609 are applied to suppress EMI, enhance ESD protection and facilitate debugging. It is recommended to use serial resistors, and mount 0 Ω resistors by default.
 4. It is recommended to take electrostatic discharge (ESD) protection measures near the (U)SIM card connector. The TVS array with parasitic capacitance not exceeding 15 pF must be placed as close as possible to the (U)SIM card connector.
 5. USIM1_DATA MUST be connected to USIM1_VDD with a 10 k Ω pull-up resistor (R0605). It can improve anti-jamming capability when long layout trace and sensitive environments are applied, and should be placed close to the (U)SIM card connector.
 6. It is recommended to add 33 pF capacitors on the traces of USIM1_DATA, USIM1_CLK and USIM1_RST to filter out RF interference. Note that the capacitance value must not be too large; otherwise, it will cause the signal waveform to slow down, fail to filter out RF interference, and may even result in card reading failure.
 7. (U)SIM2 interface has the same design requirements as (U)SIM1 interface.

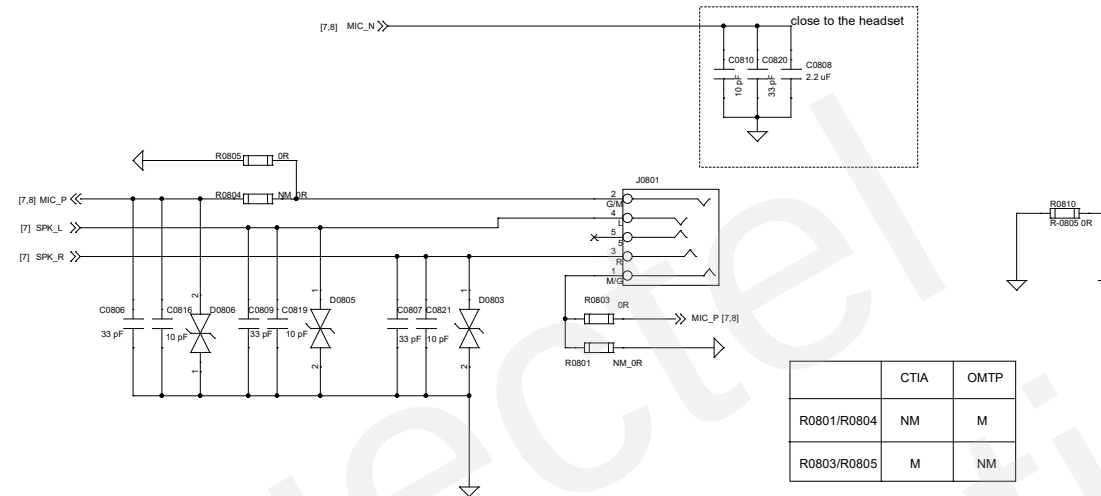
Audio Codec Design (Part 1)



NOTE:

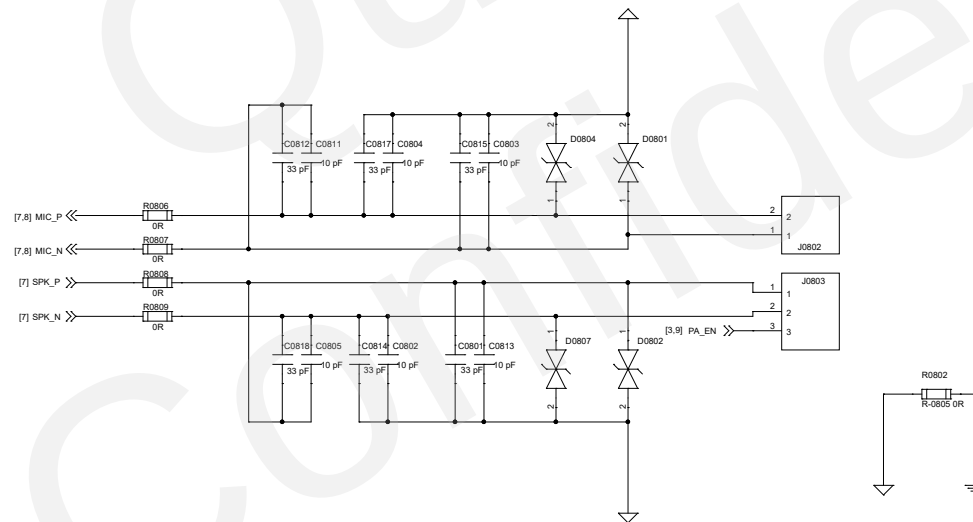
1. ALC5616 power-up sequence: DBVDD/I2C pull-up power/AVDD/DACREF/CPVDD → MICVDD → Software initialization.
2. ALC5616 power-down sequence: disable codec function by software → MICVDD → DBVDD/I2C pull-up power/AVDD/DACREF/CPVDD.
3. The pull-up resistors (R0705 and R0710) are recommended to be reserved and not mounted by default. The I2C interface of the codec integrates pull-up resistors internally, so no external pull-ups are needed.
In case of signal quality issues caused by excessive routing length of the I2C interface, external pull-ups can be added to enhance performance.
4. The module will automatically initialize the codec via I2C interface after it is turned on successfully, so all power supplies for the codec need to be turned on before codec initialization.
5. Keep the codec away from interference sources such as RF and power supplies, and surround the codec audio signal with ground as much as possible.

Audio - Headset Application



	CTIA	OMTP
R0801/R0804	NM	M
R0803/R0805	M	NM

Audio - Loudspeaker Application



NOTE:

1. For larger power loads such as loudspeaker, an audio power amplifier should be added in the design.
2. The maximum capacitive loading for loudspeaker is 330 pF and the maximum capacitive loading for microphone is 250 pF.
3. The analog GND should be connected to the main GND via the 0 Ω resistor (R0802).
4. J0803 is a simplified loudspeaker output circuit of an external power amplifier. To suppress pop, it is recommended to control the external power amplifier through PA_EN, enabling it to be turned on after the module's audio function is established.
5. In headset applications, the MIC and SPK signals need to be routed as differential pairs.
6. All MIC and SPK signals traces should be surrounded with GND on the layer and ground planes above and below, and far away from noises such as clock and DC-DC signals.

Audio - Loudspeaker Application

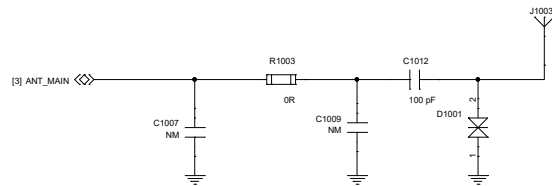


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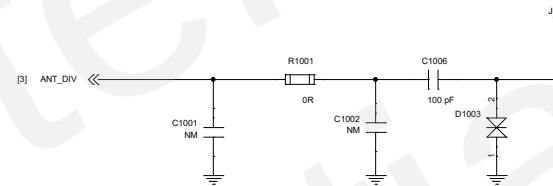
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Antenna Interface Design

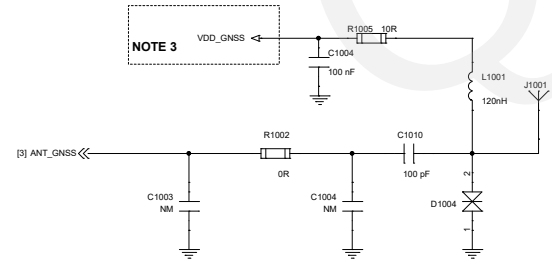
Main Antenna Interface



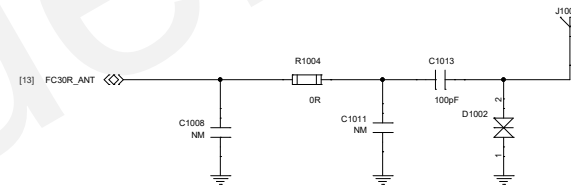
Rx-diversity Antenna Interface



GNSS Antenna Interface



Wi-Fi Module (FC30R) Antenna Interface



NOTE:

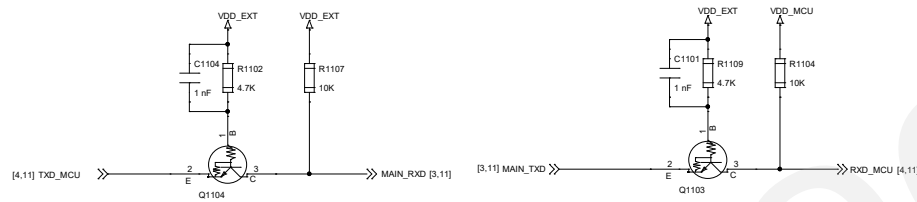
1. The main and diversity antenna circuits must be reserved. Dual L-type circuits are recommended to be used for all antenna circuits to facilitate future debugging. The single-end impedance of the RF antenna is 50 Ω .
2. The single-end impedance of RF antennas is 50 Ω .
3. The voltage of VDD_GNSS depends on the actual situation.
4. ESD protection components need to be added close to the antennas to effectively prevent static electricity. Junction capacitance of ESD protection components on the antenna interfaces is recommended to be 0.05 pF.
5. C1006, C1010, C1012 and C1013 are used for DC-blocking and the capacitance value is recommended to be 100 pF.

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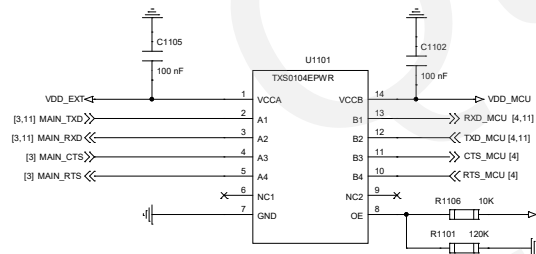
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UART & USB Interface Design

Level-shifting Circuit - Transistor Solution



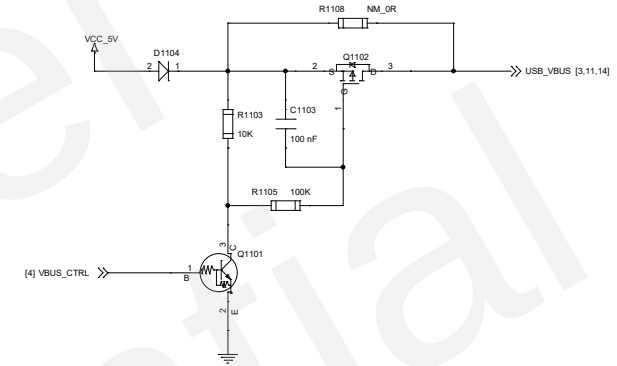
Level-shifting Circuit - IC Solution (Recommended)



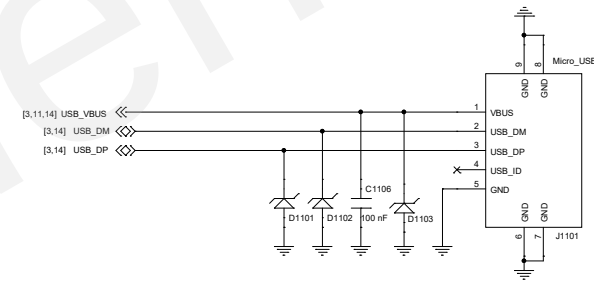
NOTE:

- There are two level-shifting solutions: transistor solution and IC solution, and the latter one is recommended.
- The power supply of TXS0104EPWR's VCCA should not exceed that of VCCB. For more details, please refer to the datasheet of TXS0104EPWR.
- The transistor solution is not applicable for applications with baud rates exceeding 460 kbps, and the capacitors C1101 and C1104 of 1 nF can improve signal quality.

USB_VBUS Control



USB 2.0 Connector



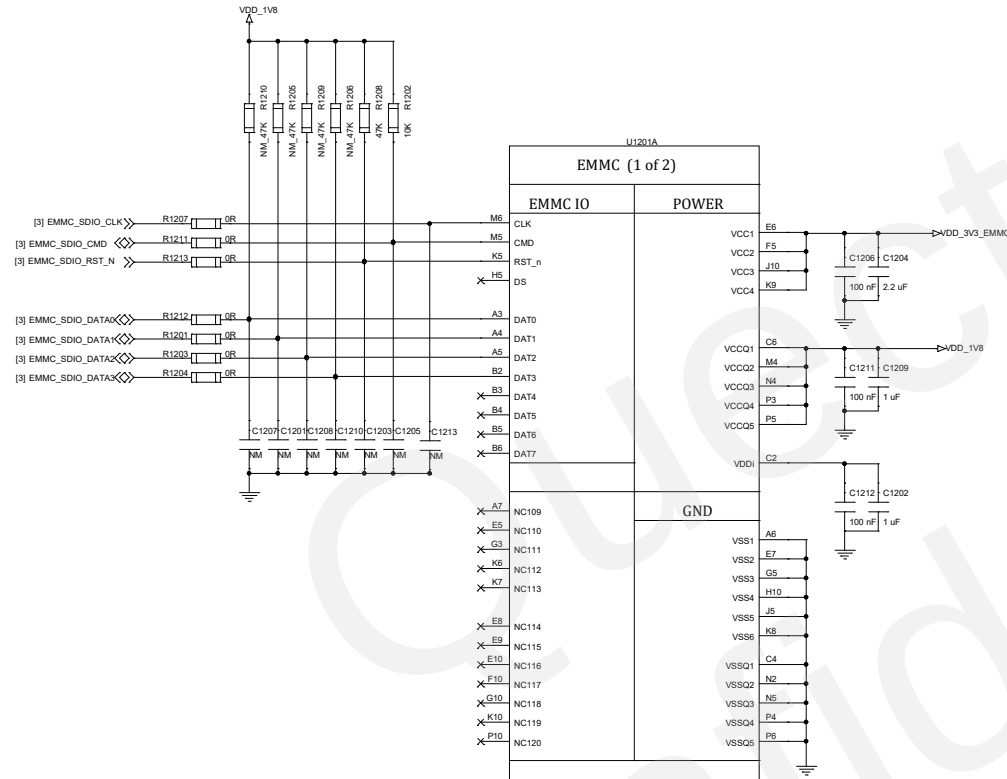
NOTE:

- The USB interface supports USB 2.0 high-speed mode and full-speed mode. It requires the MCU to support USB host mode.
- USB_VBUS is powered by this MOSFET circuit when the module is connected to MCU, and powered by PC when it is connected to PC. The USB_VBUS is used for detecting USB connection.
- VBUS_CTRL is used to turn on/off the power supply of USB_VBUS when the module connects to MCU.
- The parasitic capacitance of the ESD protection device on the USB should be less than 2 pF.
- D1104 can prevent USB_VBUS backflow when the module connects to PC via USB.

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eMMC Application Design



NOTE:

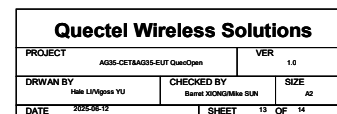
1. Different eMMC models may vary in requirements for capacitor parameters, which can be designed according to relevant specifications.
2. For detailed information of eMMC, refer to the corresponding device specification.

U1201B	
EMMC (2 of 2)	
NC	NC
X A1	NC1
X A2	NC2
X A8	NC3
X A9	NC4
X A10	NC5
X A11	NC6
X A12	NC7
X A13	NC8
X A14	NC9
X B1	NC10
X B7	NC11
X B8	NC12
X B9	NC13
X B10	NC14
X B11	NC15
X B12	NC16
X B13	NC17
X B14	NC18
X C1	NC19
X C3	NC20
X C5	NC21
X C7	NC22
X C8	NC23
X C9	NC24
X C10	NC25
X C11	NC26
X C12	NC27
X C13	NC28
X C14	NC29
X D1	NC30
X D2	NC31
X D3	NC32
X D4	NC33
X D12	NC34
X D13	NC35
X D14	NC36
X E1	NC37
X E2	NC38
X E3	NC39
X E12	NC40
X E13	NC41
X E14	NC42
X F1	NC43
X F2	NC44
X F3	NC45
X F12	NC46
X F13	NC47
X F14	NC48
X G1	NC49
X G2	NC50
X G12	NC51
X G13	NC52
X G14	NC53
X H1	NC54
X H9	NC55
X H5	NC56
X H12	NC57
X H13	NC58
X H14	NC59
X J1	NC60
X J2	NC61
X J3	NC62
X J12	NC63
X J13	NC64
X J14	NC65
X K1	NC66
X K2	NC67
X K3	NC68
X K12	NC69
X K13	NC70
X K14	NC71
X L1	NC72
X L2	NC73
X L3	NC74
X L12	NC75
X L13	NC76
X L14	NC77
X M1	NC78
X M2	NC79
X M3	NC80
X M7	NC81
X M8	NC82
X M9	NC83
X M10	NC84
X M11	NC85
X M12	NC86
X M13	NC87
X M14	NC88
X N1	NC89
X N3	NC90
X N8	NC91
X N7	NC92
X N8	NC93
X N8	NC94
X N10	NC95
X N11	NC96
X N12	NC97
X N13	NC98
X N14	NC99
X P1	NC100
X P2	NC101
X P7	NC102
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X P9	NC104
X P11	NC105
X P12	NC106
X P13	NC107
X P14	NC108

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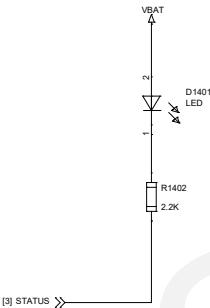
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VOUT = $1.207 \times (1 + R1315 / R1319) \approx 1.8 \text{ V}$



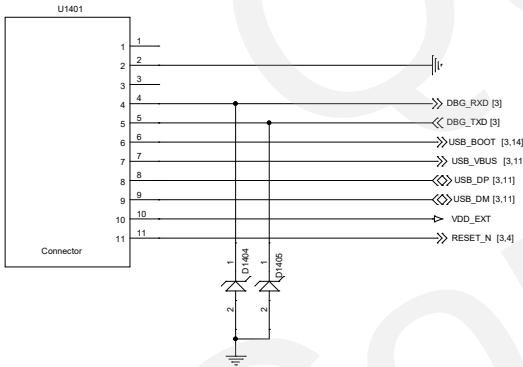
Other Designs

Indicator



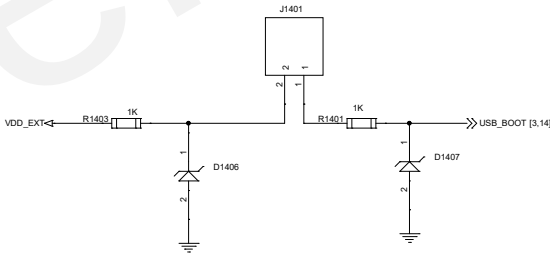
- NOTE:**
- 1. The STATUS is an open drain output pin.
 - 2. If the low current consumption is required when your device is in sleep mode, replace the power supply VBAT of the STATUS indicator with an external controllable one, which can be turned off when the module is in low power mode to reduce the power consumption.

Reserved Test Points



- NOTE:**
- 1. Test points for both USB and debug UART interfaces are reserved for capturing logs.
 - 2. The debug UART interface supports 1.8 V power domain, and a voltage-level translator should be used if the power domain of your application is 3.3 V.
 - 3. The parasitic capacitance of ESD protection components on USB data traces should not exceed 2 pF.

Forced Download Interface



- NOTE:**
- 1. It is recommended to reserve the USB_BOOT interface design.
 - 2. USB_BOOT is kept open by default. Pull up USB_BOOT to VDD_EXT before turning on the module, and then the module will enter forced download mode.